

Ferns & Petals Sales Analysis Dashboard

Project Report

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Role: Aspiring Data Analyst

Tool Used: Microsoft Excel Online

Project Type: End-to-End Data Analytics Dashboard

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1. Executive Summary

This project presents an end-to-end sales analysis of Ferns & Petals (FNP) using Microsoft Excel Online. The objective was to transform raw sales data into meaningful business insights through data cleaning, data modeling, KPI development, interactive dashboards, and visualization techniques.

The dashboard enables stakeholders to monitor sales performance, analyze customer purchasing behavior, identify high-performing products and occasions, evaluate delivery efficiency, and support data-driven business decisions.

The project demonstrates practical Excel skills commonly required for Data Analyst roles, including data transformation, lookup functions, Pivot Tables, Pivot Charts, KPI calculations, and dashboard design.

2. Business Problem

Ferns & Petals operates in the online gifting industry, offering products such as flowers, cakes, chocolates, soft toys, and personalized gifts.

As the business grows, management requires an interactive reporting solution to answer critical business questions such as:

- Which months generate the highest revenue?
- Which occasions contribute most to sales?
- Which products perform best?
- Which cities generate the highest number of orders?
- Who are the most valuable customers?
- How efficient is the delivery process?

Without a centralized dashboard, answering these questions requires manually reviewing thousands of records, making decision-making slow and inefficient.

3. Project Objectives

The objectives of this project were to:

- Clean and prepare raw sales data.
 - Build a master dataset by integrating multiple tables.
 - Perform business analysis using Pivot Tables.
 - Create meaningful Key Performance Indicators (KPIs).
 - Design an interactive executive dashboard.
 - Generate actionable business insights and recommendations.
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4. Dataset Overview

The project uses three datasets representing different aspects of the business.

Orders Dataset

Contains transactional information including:

- Order ID
 - Customer ID
 - Product ID
 - Quantity
 - Order Date
 - Delivery Date
 - Order Time
 - Delivery Time
 - Location
 - Occasion
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Customers Dataset

Contains customer-related information including:

- Customer ID

- Customer Name
 - City
 - Contact Number
 - Email
 - Gender
 - Address
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Products Dataset

Contains product information including:

- Product ID
 - Product Name
 - Category
 - Price
 - Occasion
 - Product Description
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5. Data Preparation

Before analysis, the datasets were cleaned and transformed to ensure accuracy and consistency.

The following preprocessing tasks were performed:

- Checked for missing values.
 - Standardized date formats.
 - Validated numeric fields.
 - Verified Customer IDs and Product IDs.
 - Created relationships between datasets using XLOOKUP.
 - Calculated additional business metrics.
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6. Data Modeling

The three datasets were combined into a single Master Data table using XLOOKUP.

Additional calculated fields were created, including:

- Customer Name

- Product Name
- Product Category
- Revenue
- Delivery Days
- Month
- Quarter
- Weekday

These derived fields simplified business analysis and dashboard creation.

7. Key Performance Indicators (KPIs)

The dashboard includes the following KPIs:

- Total Revenue
- Total Orders
- Total Customers
- Average Order Value
- Average Delivery Time

These KPIs provide a quick overview of business performance.

8. Dashboard Features

The interactive dashboard includes:

Revenue Analysis

- Monthly Revenue Trend
- Revenue by Occasion

Product Analysis

- Top 10 Products
- Revenue by Category

Customer Analysis

- Top 10 Customers

Geographic Analysis

- Top 10 Cities

Performance Metrics

- KPI Cards
 - Interactive Slicers
 - Dynamic Pivot Charts
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9. Excel Skills Demonstrated

This project demonstrates proficiency in:

Data Cleaning

- Removing inconsistencies
- Data validation
- Formatting

Data Transformation

- XLOOKUP
- Calculated columns
- Revenue calculations

Data Analysis

- Pivot Tables
- Pivot Charts
- Sorting
- Filtering
- Top N Analysis

Dashboard Development

- KPI Cards
 - Interactive Slicers
 - Executive Dashboard Design
 - Business Reporting
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10. Key Business Insights

(Replace the placeholders below with your actual findings from the dashboard.)

- ₹[Revenue] total revenue was generated during the analysis period.
 - [Month] recorded the highest monthly revenue.
 - [Occasion] contributed the highest sales among all occasions.
 - [Product] emerged as the highest revenue-generating product.
 - [Category] generated the largest share of product revenue.
 - [City] recorded the highest number of customer orders.
 - The average delivery time was [X] days.
 - The Top 10 customers contributed a significant percentage of total revenue.
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11. Business Recommendations

Based on the analysis, the following recommendations are proposed:

Increase Inventory

Increase stock availability before high-demand occasions such as Valentine's Day and Raksha Bandhan.

Improve Marketing

Launch promotional campaigns during low-performing months to improve seasonal revenue.

Reward Loyal Customers

Introduce customer loyalty programs targeting high-value customers.

Optimize Product Mix

Promote low-performing categories through bundled offers and seasonal discounts.

Improve Delivery Efficiency

Monitor logistics performance to reduce delivery time and improve customer satisfaction.

12. Project Challenges

Several challenges were encountered during the project:

- Integrating multiple datasets into a unified master table.
- Designing an interactive dashboard using Excel Online.
- Managing Pivot Tables and slicers within Excel Online's limitations.
- Ensuring consistent formatting across charts and KPIs.

These challenges were addressed using structured data preparation techniques and efficient dashboard design.

13. Learning Outcomes

This project strengthened practical skills in:

- Business Intelligence
 - Data Cleaning
 - Data Visualization
 - Dashboard Development
 - Excel Data Analysis
 - KPI Design
 - Business Reporting
 - Analytical Thinking
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14. Conclusion

The Ferns & Petals Sales Analysis Dashboard successfully transforms raw transactional data into an interactive business intelligence solution.

The dashboard provides management with a centralized view of business performance, enabling faster and more informed decision-making. Through effective use of Microsoft Excel Online, the

project demonstrates that advanced analytical reporting can be achieved without specialized BI software.

This project reflects an end-to-end data analytics workflow—from raw data preparation and modeling to visualization, interpretation, and strategic recommendations.

□ Appendix

Tools Used

- Microsoft Excel Online
- Git
- GitHub

Excel Features Used

- Tables
- XLOOKUP
- Pivot Tables
- Pivot Charts
- Slicers
- Conditional Formatting
- Data Validation
- Custom Number Formatting
- Sorting & Filtering

Files Included

- FNP_Sales_Dashboard.xlsx
- Orders.csv
- Customers.csv
- Products.csv
- README.md
- Project_Report.pdf